

Final Surge OAuth2 Token Acquisition

Access Request

Endpoint: GET <https://log.finalsurge.com/oauth/authorize>

Parameters

client-id	Your Applications Client ID
redirect-uri	URL to which the user will be redirected with the authorization code. This must be to the callback domain associated with the application or its sub-domain. <i>"localhost" and "127.0.0.1" are white-listed</i>
state	Returned to your application in the redirect URI

After the user authorizes or denies your application the client will be redirected to the redirect-uri.

If the user authorized your client a new variable will be available in the querystring called "code." Otherwise a new variable will be included called "error" with the value "access_denied."

Acquiring the User's Token

Endpoint: POST <https://log.finalsurge.com/oauth/token>

Parameters:

client-id	Your Applications Client ID
client-secret	Your Applications Client Secret
code	The code received from a the authorization step

Response

```
{
  "access_token": "bfe2fbd7-8862-4d85-9839-3ed70abcef11",
  "athlete": {
    "id": "16238aab-bd1d-4a89-9b17-50f33630a007",
    "firstname": "Brian",
    "lastname": "Roberds"
  },
  "error": null
}
```

Final Surge Partner API Calls

Call Authorization Requirements

Base Endpoint: <https://log.finalsurge.com>

Client Authorized Call Required Headers

client-id : <provided>

client-secret : <provided>

User Authorized Call Required Headers

client-id : <provided>

Authorization : Bearer {{token}}

Invalid Token Response

```
{  
  "Success": false,  
  "ErrorNumber": 401,  
  "ErrorDescription": "Access Forbidden"  
}
```

POST /API/v1/LoginToken

User Authorized

Generates a Token to use as an auto login to the Final Surge Beta System. Use the provided token in the response to redirect the client's browser to: <https://beta.finalsurge.com/LoginWithToken?t=<token>>

**This token will expire within 10 seconds of generation*

Parameters

none

Response

```
{
  "Success": true,
  "ErrorNumber": null,
  "ErrorDescription": null,
  "Token": "<token>"
}
```

GET /API/v1/ProfileInfo

User Authorized

Read profile data saved for this app for the user account within the Final Surge System

Parameters

none

Response

```
{
  "uniqueid":<value>,
  "profile":<value>,
  "Success": true,
  "ErrorNumber": null,
  "ErrorDescription": null
}
```

POST /API/v1/ProfileInfo

User Authorized

Allows your application to save your unique identifier to the user account within the Final Surge System as well as profile information

Parameters

uniqueid (string – max length: 200)	Your unique identifier for this user
profile (string – max length: 3000)	Additional profile information about this user

Response

```
{  
  "Success": true,  
  "ErrorNumber": null,  
  "ErrorDescription": null  
}
```

POST /API/v1/uploads

User Authorized

We currently support Fit and TCX files. **The only required parameter is the file itself**

Parameters

(multipart/form-data)

file	Encoded File Data
workout-id (string – max length: 50)	Your application's unique identifier for this workout
workout-title (string – max length: 200)	Title for the workout <i>*ignored if merging with an existing workout that has a title</i>
workout-description (string – max length: 3000)	Description for the workout <i>*ignored if merging with an existing workout that has a description</i>
workout-notes (string – max length: 3000)	User's notes for this workout <i>*Will be stored as "Post Workout Notes" within Final Surge</i>

Response

You will receive an Http Status Code of 201 if the file is successfully queued for processing

GET /API/v1/UpcomingWorkouts

User Authorized

Retrieves the next 5 workouts for a user including the users current date. This will also send completed workouts. Both parameters are taken into account. If neither are provided the five next workouts are returned.

**Only workouts from the upcoming 14 days are possible to return*

Parameters

NumDays (int) <i>optional</i>	Number of days who's workouts should be returned – Minimum 1, Maximum 7. These days include the user's current day
NumWorkouts (int) <i>optional</i>	Number workouts to return – Minimum 1, Maximum 21

Response

```
{
  "Success": true,
  "ErrorNumber": null,
  "ErrorMessage": null,
  "Workouts": [
    {
      "WorkoutURL": "https://log.finalsurge.com/WorkoutDetails?s=f73de4d9-878c-4b6e-af37-ab2cf3540a28&id=cc3b49d9-0b25-44ff-9799-cdb5446f2d3a",
      "WorkoutDate": "2019-01-24T00:00:00",
      "WorkoutTime": "17:00:00",
      "WorkoutCode": "TRLTP",
      "WorkoutTitle": "Tempo Run",
      "WorkoutDescription": "10-15 minute Warm-Up + Tempo Run: 10-15 minutes + 10-15 minute Cool-down\r\n\r\nWorkout Purpose: \r\nBuild stamina - lactate threshold pace.",
      "WorkoutTypeName": "Run",
      "WorkoutSubTypeName": "Tempo Run",
      "WorkoutCompleted": false,
      "WorkoutIcon": 1,
      "PlannedTime": 1800,
      "PlannedDistance": 25.5,
      "PlannedDistanceType": "mi",
      "PlannedPace": "5:40-5:55",
      "PlannedPaceType": "min/mi",
      "StructuredWorkoutURLs": {
        "fit": null,
        "json_garmin_v1": null,
        "json_fs_v1": null,
        "mrc": null,
        "zwo": null
      }
    }
  ]
}
```

All workout fields are nullable except for WorkoutDate, WorkoutCompleted, WorkoutTypeName and WorkoutIcon

WorkoutIcon Values: (If you choose to implement icons)

1. Run
2. Bike
3. Swim
4. Cross Training
5. Strength Training
6. Rest Day
7. Recovery
8. Other
9. Transition
10. Custom
11. Walk

The collection StructuredWorkoutURLs returns different file types for the structured workout that are supported. To download those files you must include the user authorization headers on the request (client-id & Authorization)

fit	Industry standard fit file with structured workout steps loaded
json_garmin_v1	A json object tailored to Garmin's Training API workout specifications. This is exactly what we send into Garmin for our Structured Workout Sync
json_fs_v1	A json object with all our features from the workout builder. See the Workout Object specification below for documentation
mrc	A standard MRC file format. Duration and Power workouts only
zwo	Swift's Structured Workout format. Duration and Power workouts only

Object: Workout

This is the base workout object. See a full example of a workout after Step documentation.

workoutId	Required	dfd
workoutName	Required	The name of the workout as specified in the workout builder
sport	Required	RUNNING, CYCLING, or SWIMMING
poolLength	Conditional	<i>Swimming Workout:</i> pool length integer
poolLengthUnit	Conditional	<i>Swimming Workout:</i> pool length unit (METRIC or STATUTE)
steps	Required	Collection of Workout Steps. Type attribute denotes the step fields used. See following documentation

Type: WorkoutStep

This is the primary step within our workouts. It defines some work to do. It can be for a defined time, distance, or rely on the a interaction from the athlete to mark complete the step and move on.

intensity	Required	ACTIVE – This step is considered an aggressive part of the workout. REST – This step is usually a period in the workout where the athlete is recovering/resting WARMUP – This step is usually the first step within a workout to get the athlete ready for the bulk of the workout COOLDOWN – This step is usually the last step within a workout.
title	Optional	A quick title of this step. Great for displaying when offering a summary of the workout.
comment	Optional	Usually contains more details about this step or reminders while performing this step.
durationType	Required	TIME – A time based step DISTANCE – A bistance base step OPEN – Requires athlete interaction to complete this step and move on or mark the workout complete
durationValue	Conditional	<i>durationType: TIME</i> – The amount of time in seconds <i>durationType: DISTANCE</i> – The amount of distance in meters <i>durationType: OPEN</i> – Null value
primaryTargetType	Optional	POWER – This step has a specified wattage range. Target Values are in watts. HEART_RATE – Has a specified heart rate range. Target values are in bpm. PACE – Has a specified speed or pace range. Target values are in meters/second <i>* If null, this step has no target range to maintain</i>
primaryTargetLow	Conditional	<i>primaryTargetType is supplied</i> - Low range of watts, bpm, or pace/speed
primaryTargetHigh	Conditional	<i>primaryTargetType is supplied</i> – High range of watts, bpm, or pace/speed
secondaryTargetType	Conditional	CADENCE – This step has a specified RPM/SPM range <i>* A primary target range must be specified to have a secondary provided</i> <i>* if null, this step has no secondary target range to maintain</i>
secondaryTargetLow	Conditional	<i>secondaryTargetType is supplied</i> - Low range of rpm/spm
secondaryTargetHigh	Conditional	<i>secondaryTargetType is supplied</i> – High range of rpm/spm

Type: WorkoutRampStep

A Ramp Step is a step where the athlete is to either build up or down from one target to the other over the specified time or distance. A ramp step cannot be an OPEN duration type. For those applications that do not support Ramps, the 'steps' collection includes the ramp represented in WorkoutSteps.

steps	Required	An array of WorkoutSteps that represents the Ramp as a series of steps. These steps can be used in the place of the Ramp if the UI does not support Ramps.
intensity	Required	ACTIVE – This step is considered an aggressive part of the workout. REST – This step is usually a period in the workout where the athlete is recovering/resting WARMUP – This step is usually the first step within a workout to get the athlete ready for the bulk of the workout COOLDOWN – This step is usually the last step within a workout.
title	Optional	A quick title of this step. Great for displaying when offering a summary of the workout.
comment	Optional	Usually contains more details about this step or reminders while performing this step.
durationType	Required	TIME – A time based step DISTANCE – A distance base step <i>* OPEN is not supported for Ramps</i>
durationValue	Conditional	<i>durationType: TIME</i> – The amount of time in seconds <i>durationType: DISTANCE</i> – The amount of distance in meters
primaryTargetType	Required	POWER – This step builds from and to a specified wattage. Target values are in watts HEART_RATE – This step builds from and to a specified heart rate. Target values are in bpm. PACE – This step builds from and to a specified pace/speed. Target values are in meters/second
primaryTargetStart	Conditional	Start wattage, bpm, or pace/speed
primaryTargetFinish	Conditional	Finish wattage, bpm, or pace/speed

Type: WorkoutRepeatStep

A Workout Repeat Step simply contains an array of Steps/Ramps/Repeats that should be completed the specified number of times. It is important to accommodate the nesting of repeats.

steps	Required	An array of WorkoutSteps that represents the Ramp as a series of steps. These steps can be used in the place of the Ramp if the UI does not support Ramps.
repeatValue	Required	The number of times to repeat the Steps specified in the steps array

FS Workout Example JSON:

```
{
  "workoutId": "0000000000000",
  "workoutName": "Test Workout",
  "sport": "RUNNING",
  "poolLength": null,
  "poolLengthUnit": null,
  "steps": [
    {
      "type": "WorkoutStep",
      "repeatValue": null,
      "steps": null,
      "intensity": "WARMUP",
      "title": "Easy Warm Up",
      "comment": "Wake up your legs",
      "durationType": "TIME",
      "durationValue": 600.0,
      "durationValueType": null,
      "primaryTargetType": "POWER",
      "secondaryTargetType": "CADENCE",
      "primaryTargetValue": null,
      "secondaryTargetValue": null,
      "primaryTargetValueLow": 175.0,
      "secondaryTargetValueLow": 20.0,
      "primaryTargetValueHigh": 225.0,
      "secondaryTargetValueHigh": 50.0,
      "primaryTargetValueStart": null,
      "primaryTargetValueFinish": null,
      "primaryTargetValueType": null,
      "strokeType": null,
      "equipmentType": null,
      "exerciseCategory": null,
      "exerciseName": null,
      "weightValue": null,
      "weightDisplayUnit": null
    },
    {
      "type": "WorkoutRampStep",
      "repeatValue": null,
      "steps": [
        {
          "type": "WorkoutStep",
          "repeatValue": null,
          "steps": null,
          "intensity": "ACTIVE",
          "title": null,
          "comment": "Get ready to work",
          "durationType": "TIME",
          "durationValue": 60.0,
          "durationValueType": null,
          "primaryTargetType": "POWER",
          "secondaryTargetType": null,
          "primaryTargetValue": null,
          "secondaryTargetValue": null,
          "primaryTargetValueLow": 200.0,
          "secondaryTargetValueLow": null,
          "primaryTargetValueHigh": 225.0,
          "secondaryTargetValueHigh": null,
          "primaryTargetValueStart": null,
          "primaryTargetValueFinish": null,
          "primaryTargetValueType": null,
          "strokeType": null,
          "equipmentType": null,
          "exerciseCategory": null,
          "exerciseName": null,
          "weightValue": null,
          "weightDisplayUnit": null
        },
        {
          "type": "WorkoutStep",
          "repeatValue": null,
          "steps": null,

```

```

    "intensity": "ACTIVE",
    "title": null,
    "comment": "Get ready to work",
    "durationType": "TIME",
    "durationValue": 60.0,
    "durationValueType": null,
    "primaryTargetType": "POWER",
    "secondaryTargetType": null,
    "primaryTargetValue": null,
    "secondaryTargetValue": null,
    "primaryTargetValueLow": 225.0,
    "secondaryTargetValueLow": null,
    "primaryTargetValueHigh": 250.0,
    "secondaryTargetValueHigh": null,
    "primaryTargetValueStart": null,
    "primaryTargetValueFinish": null,
    "primaryTargetValueType": null,
    "strokeType": null,
    "equipmentType": null,
    "exerciseCategory": null,
    "exerciseName": null,
    "weightValue": null,
    "weightDisplayUnit": null
  },
  {
    "type": "WorkoutStep",
    "repeatValue": null,
    "steps": null,
    "intensity": "ACTIVE",
    "title": null,
    "comment": "Get ready to work",
    "durationType": "TIME",
    "durationValue": 60.0,
    "durationValueType": null,
    "primaryTargetType": "POWER",
    "secondaryTargetType": null,
    "primaryTargetValue": null,
    "secondaryTargetValue": null,
    "primaryTargetValueLow": 250.0,
    "secondaryTargetValueLow": null,
    "primaryTargetValueHigh": 275.0,
    "secondaryTargetValueHigh": null,
    "primaryTargetValueStart": null,
    "primaryTargetValueFinish": null,
    "primaryTargetValueType": null,
    "strokeType": null,
    "equipmentType": null,
    "exerciseCategory": null,
    "exerciseName": null,
    "weightValue": null,
    "weightDisplayUnit": null
  },
  {
    "type": "WorkoutStep",
    "repeatValue": null,
    "steps": null,
    "intensity": "ACTIVE",
    "title": null,
    "comment": "Get ready to work",
    "durationType": "TIME",
    "durationValue": 60.0,
    "durationValueType": null,
    "primaryTargetType": "POWER",
    "secondaryTargetType": null,
    "primaryTargetValue": null,
    "secondaryTargetValue": null,
    "primaryTargetValueLow": 275.0,
    "secondaryTargetValueLow": null,
    "primaryTargetValueHigh": 300.0,
    "secondaryTargetValueHigh": null,
    "primaryTargetValueStart": null,
    "primaryTargetValueFinish": null,
    "primaryTargetValueType": null,
    "strokeType": null,
  }
}

```

```

        "equipmentType": null,
        "exerciseCategory": null,
        "exerciseName": null,
        "weightValue": null,
        "weightDisplayUnit": null
    },
    ],
    "intensity": "ACTIVE",
    "title": "Build up to some speed",
    "comment": "Get ready to work",
    "durationType": "TIME",
    "durationValue": 240.0,
    "durationValueType": null,
    "primaryTargetType": "POWER",
    "secondaryTargetType": null,
    "primaryTargetValue": null,
    "secondaryTargetValue": null,
    "primaryTargetValueLow": null,
    "secondaryTargetValueLow": null,
    "primaryTargetValueHigh": null,
    "secondaryTargetValueHigh": null,
    "primaryTargetValueStart": 200.0,
    "primaryTargetValueFinish": 300.0,
    "primaryTargetValueType": null,
    "strokeType": null,
    "equipmentType": null,
    "exerciseCategory": null,
    "exerciseName": null,
    "weightValue": null,
    "weightDisplayUnit": null
},
{
    "type": "WorkoutRepeatStep",
    "repeatValue": 2.0,
    "steps": [
        {
            "type": "WorkoutStep",
            "repeatValue": null,
            "steps": null,
            "intensity": "ACTIVE",
            "title": "Run Hard",
            "comment": "But not too hard",
            "durationType": "TIME",
            "durationValue": 120.0,
            "durationValueType": null,
            "primaryTargetType": "POWER",
            "secondaryTargetType": null,
            "primaryTargetValue": null,
            "secondaryTargetValue": null,
            "primaryTargetValueLow": 350.0,
            "secondaryTargetValueLow": null,
            "primaryTargetValueHigh": 400.0,
            "secondaryTargetValueHigh": null,
            "primaryTargetValueStart": null,
            "primaryTargetValueFinish": null,
            "primaryTargetValueType": null,
            "strokeType": null,
            "equipmentType": null,
            "exerciseCategory": null,
            "exerciseName": null,
            "weightValue": null,
            "weightDisplayUnit": null
        },
        {
            "type": "WorkoutStep",
            "repeatValue": null,
            "steps": null,
            "intensity": "REST",
            "title": "Quick Break",
            "comment": "Keep moving though",
            "durationType": "TIME",
            "durationValue": 30.0,
            "durationValueType": null,
            "primaryTargetType": "OPEN",

```

```

    "secondaryTargetType": null,
    "primaryTargetValue": null,
    "secondaryTargetValue": null,
    "primaryTargetValueLow": 0.0,
    "secondaryTargetValueLow": null,
    "primaryTargetValueHigh": 0.0,
    "secondaryTargetValueHigh": null,
    "primaryTargetValueStart": null,
    "primaryTargetValueFinish": null,
    "primaryTargetValueType": null,
    "strokeType": null,
    "equipmentType": null,
    "exerciseCategory": null,
    "exerciseName": null,
    "weightValue": null,
    "weightDisplayUnit": null
  },
  {
    "type": "WorkoutRepeatStep",
    "repeatValue": 5.0,
    "steps": [
      {
        "type": "WorkoutStep",
        "repeatValue": null,
        "steps": null,
        "intensity": "ACTIVE",
        "title": "Run Super Hard",
        "comment": "This is going to burn!",
        "durationType": "TIME",
        "durationValue": 60.0,
        "durationValueType": null,
        "primaryTargetType": "POWER",
        "secondaryTargetType": null,
        "primaryTargetValue": null,
        "secondaryTargetValue": null,
        "primaryTargetValueLow": 400.0,
        "secondaryTargetValueLow": null,
        "primaryTargetValueHigh": 450.0,
        "secondaryTargetValueHigh": null,
        "primaryTargetValueStart": null,
        "primaryTargetValueFinish": null,
        "primaryTargetValueType": null,
        "strokeType": null,
        "equipmentType": null,
        "exerciseCategory": null,
        "exerciseName": null,
        "weightValue": null,
        "weightDisplayUnit": null
      },
      {
        "type": "WorkoutStep",
        "repeatValue": null,
        "steps": null,
        "intensity": "REST",
        "title": "Another Quick Break",
        "comment": "Remember to Keep moving",
        "durationType": "TIME",
        "durationValue": 30.0,
        "durationValueType": null,
        "primaryTargetType": "OPEN",
        "secondaryTargetType": null,
        "primaryTargetValue": null,
        "secondaryTargetValue": null,
        "primaryTargetValueLow": 0.0,
        "secondaryTargetValueLow": null,
        "primaryTargetValueHigh": 0.0,
        "secondaryTargetValueHigh": null,
        "primaryTargetValueStart": null,
        "primaryTargetValueFinish": null,
        "primaryTargetValueType": null,
        "strokeType": null,
        "equipmentType": null,
        "exerciseCategory": null,
        "exerciseName": null,

```



```

        "weightValue": null,
        "weightDisplayUnit": null
    }
],
    "intensity": null,
    "title": null,
    "comment": null,
    "durationType": null,
    "durationValue": null,
    "durationValueType": null,
    "primaryTargetType": null,
    "secondaryTargetType": null,
    "primaryTargetValue": null,
    "secondaryTargetValue": null,
    "primaryTargetValueLow": null,
    "secondaryTargetValueLow": null,
    "primaryTargetValueHigh": null,
    "secondaryTargetValueHigh": null,
    "primaryTargetValueStart": null,
    "primaryTargetValueFinish": null,
    "primaryTargetValueType": null,
    "strokeType": null,
    "equipmentType": null,
    "exerciseCategory": null,
    "exerciseName": null,
    "weightValue": null,
    "weightDisplayUnit": null
}
],
    "intensity": null,
    "title": null,
    "comment": null,
    "durationType": null,
    "durationValue": null,
    "durationValueType": null,
    "primaryTargetType": null,
    "secondaryTargetType": null,
    "primaryTargetValue": null,
    "secondaryTargetValue": null,
    "primaryTargetValueLow": null,
    "secondaryTargetValueLow": null,
    "primaryTargetValueHigh": null,
    "secondaryTargetValueHigh": null,
    "primaryTargetValueStart": null,
    "primaryTargetValueFinish": null,
    "primaryTargetValueType": null,
    "strokeType": null,
    "equipmentType": null,
    "exerciseCategory": null,
    "exerciseName": null,
    "weightValue": null,
    "weightDisplayUnit": null
},
{
    "type": "WorkoutStep",
    "repeatValue": null,
    "steps": null,
    "intensity": "COOLDOWN",
    "title": "Easy shake out",
    "comment": "Way to go!",
    "durationType": "TIME",
    "durationValue": 600.0,
    "durationValueType": null,
    "primaryTargetType": "POWER",
    "secondaryTargetType": null,
    "primaryTargetValue": null,
    "secondaryTargetValue": null,
    "primaryTargetValueLow": 175.0,
    "secondaryTargetValueLow": null,
    "primaryTargetValueHigh": 175.0,
    "secondaryTargetValueHigh": null,
    "primaryTargetValueStart": null,
    "primaryTargetValueFinish": null,
    "primaryTargetValueType": null,

```

```
    "strokeType": null,  
    "equipmentType": null,  
    "exerciseCategory": null,  
    "exerciseName": null,  
    "weightvalue": null,  
    "weightDisplayUnit": null  
  }  
]  
}
```